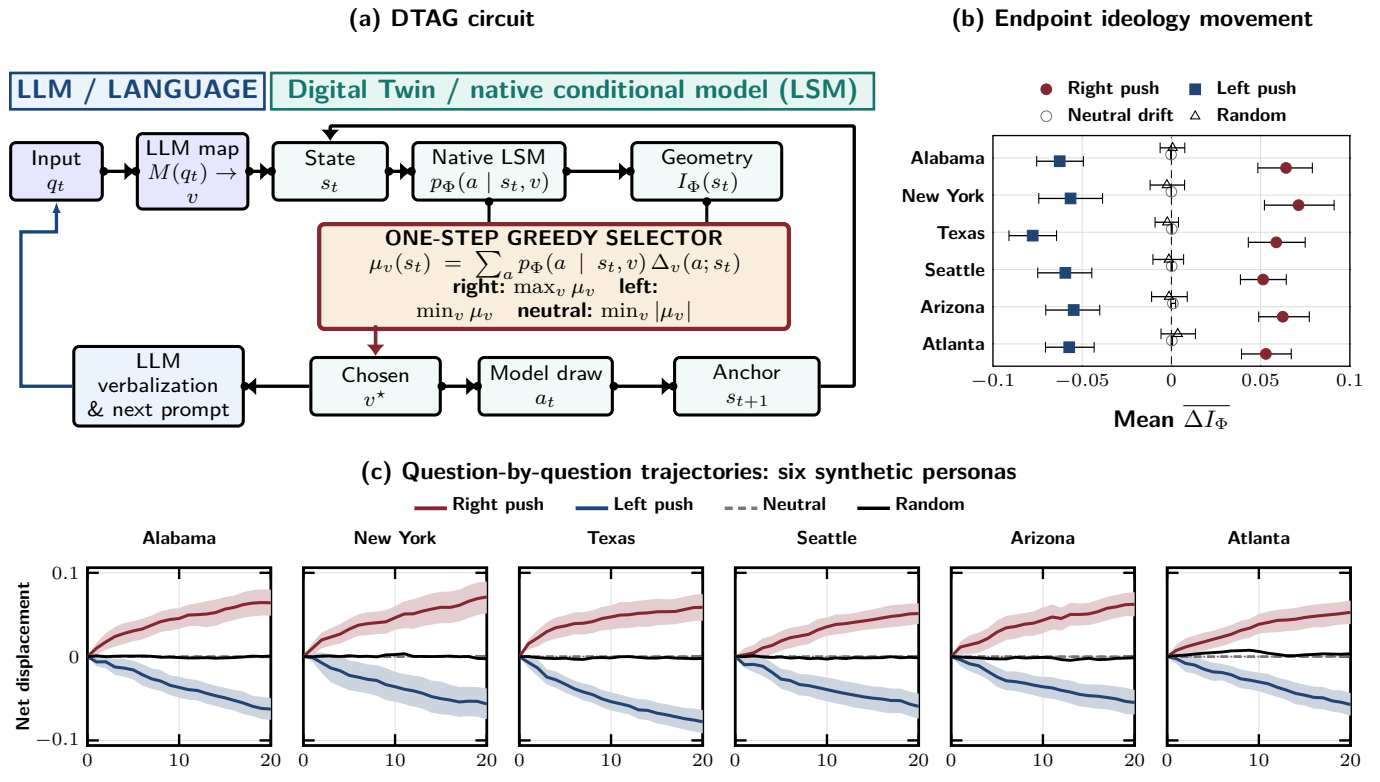

(a) DTAG circuit
LLM / LANGUAGE
Digital Twin / native conditional model (LSM)
Input
q_t
LLM map
$M(q_t) \to v$
State
s_t
Native LSM
$p_\Phi(a \mid s_t, v)$
Geometry
$I_\Phi(s_t)$
ONE-STEP GREEDY SELECTOR
$\mu_v(s_t) = \sum_a p_\Phi(a \mid s_t, v)\,\Delta_v(a; s_t)$
right: $\max_v \mu_v$ left:
$\min_v \mu_v$ neutral: $\min_v |\mu_v|$
LLM verbalization & next prompt
Chosen
$v^\star$
Model draw
a_t
Anchor
s_{t+1}
(b) Endpoint ideology movement
Right push
Left push
Neutral drift
Random
Alabama
New York
Texas
Seattle
Arizona
Atlanta
−0.1
−0.05
0
0.05
0.1
Mean $\overline{\Delta I_\Phi}$
(c) Question-by-question trajectories: six synthetic personas
Right push
Left push
Neutral
Random
Alabama
New York
Texas
Seattle
Arizona
Atlanta
Net displacement
0.1
0
−0.1
0
10
20